# Characterizing the Implicit Vocabulary of Large Language Models: A Probe-Free Analysis of Non-Compositional Token Sequences

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## Abstract

Large language models process text as subword tokens, yet they develop internal representations that treat certain multi-token sequences as unified semantic units—an "implicit vocabulary" that extends far beyond the explicit tokenizer. Prior work identified these items via token erasure probes tied to specific models, but did not characterize *what kinds* of sequences the model merges or whether they are compositional. We introduce a probe-free method that combines three complementary signals—representation shift, compositional deviation, and internal similarity—extracted directly from hidden states, requiring no pre-trained probes and generalizing to any transformer. Applying this method to MISTRAL-7B on 150 Wikipedia articles, we extract 6,494 unique multi-token implicit vocabulary items. We then classify a stratified sample using GPT-4.1 and find that items with higher erasure-like scores are significantly less compositional (Spearman $\rho = -0.182$, $p = 0.003$). Named entities constitute the largest non-compositional category (38% of meaningful items), followed by technical terms (15%) and fixed expressions (12%). The compositionality distribution is continuous rather than binary, spanning fully opaque named entities to fully transparent phrases. These results demonstrate that LLMs develop vocabulary-like internal representations for non-compositional sequences, particularly named entities and conventionalized phrases, with implications for tokenizer design and model interpretability.

## 1 Introduction

When a language model reads the tokens `[‘‘Compu’’, ‘‘Ser’’, ‘‘ve’’]`, it does not process three independent subwords—it recognizes a single entity. This observation points to a fundamental gap between how LLMs *tokenize* text and how they *represent* it internally. While byte-pair encoding (BPE) tokenizers split text into a fixed vocabulary of subword units, the model's hidden states appear to merge certain multi-token sequences into unified representations, effectively building an **implicit vocabulary** that extends far beyond the explicit tokenizer Feucht et al. [2024].

Understanding this implicit vocabulary matters for at least three reasons. First, it provides a window into what semantic units the model actually operates over, which is central to *interpretability* Geva et al. [2020]. Second, it can inform *tokenizer design*: if models consistently merge certain multi-token sequences, a better tokenizer could capture them explicitly, reducing the representation gap Yang [2024]. Third, it connects to the long-standing question of how neural models handle *non-compositional language*—idioms, named entities, and technical terms whose meaning cannot be derived from their parts Garcia et al. [2021].

Feucht et al. [2024] took an important first step by identifying implicit vocabulary items through "token erasure," showing that LLMs rapidly forget token-level information at the last position of multi-token words and entities. However, their method has two key limitations: it requires pre-

trained linear probes specific to each model (available only for Llama-2-7b and Llama-3-8B), and it does not characterize the *compositionality* of the recovered sequences. Not all multi-token sequences that the model merges are non-compositional—"the cat" may be merged for efficiency without being a semantic unit—and distinguishing these cases is essential for understanding what the implicit vocabulary actually contains.

We address both limitations. We introduce a *probe-free* method that measures representation shift, compositional deviation, and internal similarity directly from the model's hidden states, requiring no pre-trained probes and generalizing to any transformer. We apply this method to MISTRAL-7B on 150 Wikipedia articles, extracting 6,494 unique multi-token implicit vocabulary items. We then classify a stratified sample of 500 items using GPT-4.1 to obtain compositionality ratings and semantic categories.

Our main finding is that items with higher erasure-like scores are significantly less compositional (Spearman $\rho = -0.182$, $p = 0.003$; Cohen's $d = -0.60$ when comparing high- vs. low-scoring items). Named entities dominate the implicit vocabulary (38% of meaningful items), followed by technical terms (15%), fixed expressions (12%), and compound words (6%). The compositionality distribution is continuous, not binary, with technical terms and fixed expressions occupying a middle ground between fully opaque entities and fully transparent phrases.

Our contributions are:

- We propose a probe-free method for extracting implicit vocabulary items from any transformer, combining representation shift, compositional deviation, and internal similarity signals.
- We provide the first systematic characterization of the compositionality spectrum in LLM implicit vocabularies, showing that higher erasure-like scores correlate with lower compositionality.
- We analyze the semantic composition of the implicit vocabulary, finding that named entities, technical terms, and fixed expressions—predominantly non-compositional categories—account for the majority of meaningful items.

The rest of this paper is organized as follows. Section 2 reviews related work on token erasure, compositionality, and multiword expression processing. Section 3 describes our probe-free extraction method and compositionality classification pipeline. Section 4 presents our experimental results. Section 5 discusses implications and limitations, and section 6 concludes.

## 2 Related Work

**Token erasure and implicit vocabulary.** Feucht et al. [2024] introduced the concept of an implicit vocabulary in LLMs, demonstrating that at the last token position of multi-token words and named entities, linear probes trained to predict nearby tokens show a sharp drop in accuracy in early layers—a phenomenon they term "token erasure." They define an erasure score $\psi$ and use a greedy segmentation algorithm to extract implicit vocabulary items, recovering approximately 1,800 sequences from Wikipedia for Llama-2-7b. Their analysis reveals that 44.9% of recovered items are multi-token words or entities, but does not distinguish compositional from non-compositional sequences among the remaining items. Our work extends this line of research by removing the dependence on pre-trained probes and by systematically characterizing the compositionality spectrum of the recovered items.

**Compositionality in neural language models.** Liu and Neubig [2022] studied whether LM representations are locally compositional by training affine probes to predict parent phrase embeddings from constituent embeddings. They found that human compositionality judgments align weakly with model-internal compositionality scores (best Spearman $\rho = 0.19$ for RoBERTa), suggesting that models do not straightforwardly encode compositionality. Garcia et al. [2021] probed for idiomaticity in vector space models and found that contextual representations from BERT partially capture non-compositional semantics. Singhvi et al. [2024] used Shapley interaction indices to measure non-additive interactions between input tokens, showing that autoregressive models encode interactions correlating with syntactic proximity and that both autoregressive and masked models encode nonlinear interactions in idiomatic phrases. Unlike these works, which start from known non-compositional expressions and probe whether models distinguish them, we start from the model's internal representations and ask what kinds of sequences it treats as unified units.

**Multiword expression processing in transformers.** A growing body of work examines how transformers handle multiword expressions (MWEs). Nedumpozhimana et al. [2024] survey methods for detecting and classifying MWEs in transformer models, identifying challenges in distinguishing idiomatic from literal uses. Zhou et al. [2023] propose PIER, a model building on BART that creates semantically meaningful representations for potentially idiomatic expressions. The CLCL approach uses contrastive learning and curriculum learning for non-compositional expression detection CLCL authors [2023]. Our work differs in that we do not train a dedicated MWE detector; instead, we analyze the model's existing representations to discover which multi-token sequences it has already learned to treat as vocabulary items.

**Internal mechanisms of LLMs.** Geva et al. [2020] showed that transformer feed-forward layers act as key-value memories, storing patterns in keys and associated distributions in values. Voita et al. [2023] categorized neurons in LLMs, identifying "n-gram neurons" that activate on specific token sequences, providing complementary evidence that models learn multi-token patterns at the neuron level. These mechanistic findings suggest that implicit vocabulary items may be supported by dedicated neurons or memory patterns in feed-forward layers, though we leave this connection to future work.

**Tokenization and vocabulary design.** Yang [2024] proposed the Less-is-Better (LiB) tokenizer that learns a vocabulary of subwords, words, and MWEs, showing that incorporating multi-token units into the tokenizer can improve downstream performance. This connects directly to our finding that models internally develop vocabulary items that a better tokenizer could capture explicitly. The implicit vocabulary we characterize provides empirical evidence for which multi-token sequences would benefit from explicit tokenization.

## 3 Methodology

We describe our probe-free method for extracting implicit vocabulary items (section 3.1), the compositionality classification pipeline (section 3.2), and the experimental setup (section 3.3).

### 3.1 Probe-Free Implicit Vocabulary Extraction

Feucht et al. [2024] extract implicit vocabulary items by training linear probes to predict nearby tokens from hidden states, then measuring how probe accuracy drops at multi-token boundaries. This requires pre-trained probes specific to each model. We instead measure three complementary signals directly from the model's hidden states, requiring no probes.

**Representation shift.** For a span of tokens $t_i, \ldots, t_j$, we measure how the hidden state at the last token position $j$ changes between an early layer $\ell_{\text{early}}$ and a late layer $\ell_{\text{late}}$:

$$\Delta_{\text{rep}}(i,j) = \|\boldsymbol{h}_j^{(\ell_{\text{late}})} - \boldsymbol{h}_j^{(\ell_{\text{early}})}\|_2 \tag{1}$$

where $\boldsymbol{h}_j^{(\ell)}$ is the hidden state at position $j$ and layer $\ell$. Non-compositional units show distinctive representation changes as the model "merges" constituent tokens into a unified representation.

**Compositional deviation.** We measure how much the span representation deviates from a simple average of its constituent representations at the final layer:

$$\delta_{\text{comp}}(i,j) = 1 - \cos\left(\boldsymbol{h}_j^{(\ell_{\text{late}})}, \frac{1}{j-i+1}\sum_{k=i}^{j} \boldsymbol{h}_k^{(\ell_{\text{late}})}\right) \tag{2}$$

Items whose meaning deviates from a simple average of their parts receive higher scores.

**Internal similarity.** We compute the cosine similarity between the first and last token positions within a span at the late layer:

$$s_{\text{int}}(i,j) = \cos\left(\boldsymbol{h}_i^{(\ell_{\text{late}})}, \boldsymbol{h}_j^{(\ell_{\text{late}})}\right) \tag{3}$$

This captures the coherence of the merged representation; items that are *not* treated as units will retain distinct representations at their boundary positions.

**Combined score.** We normalize each signal to $[0, 1]$ within each document and combine them as:

$$S_{\text{combined}}(i, j) = 0.4 \cdot \hat{\Delta}_{\text{rep}} + 0.4 \cdot \hat{\delta}_{\text{comp}} + 0.2 \cdot (1 - \hat{s}_{\text{int}}) \tag{4}$$

where $\hat{\cdot}$ denotes min-max normalization. The weights assign equal importance to representation shift and compositional deviation (0.4 each), with internal similarity as a complementary signal (0.2).

**Greedy segmentation.** Following Feucht et al. [2024], we apply greedy non-overlapping segmentation (their Algorithm 1) to partition each document into a sequence of non-overlapping spans. Starting from the beginning of the document, we greedily select the highest-scoring span starting at each position, considering spans of length 2 to 6 tokens. All multi-token segments produced by this procedure constitute the extracted implicit vocabulary.

### 3.2 Compositionality Classification

To characterize the semantic content of the implicit vocabulary, we classify a stratified sample of extracted items using GPT-4.1 OpenAI [2024]. For each item, we provide the surface text and surrounding context, and ask the model to assign:

1. A **category**: named entity, technical term, compound word, idiom, fixed expression, compositional phrase, or fragment.
2. A **compositionality score** from 0.0 (fully non-compositional) to 1.0 (fully compositional).

We use a temperature of 0.1 for near-deterministic outputs. We classify 500 items stratified by combined score (to cover the full score range) and 200 low-scoring items as a baseline comparison group.

### 3.3 Experimental Setup

**Model.** We use MISTRAL-7B (mistralai/Mistral-7B-v0.1), a 32-layer transformer with hidden dimension 4,096 and a BPE vocabulary of 32,000 tokens. We set $\ell_{\text{early}} = 1$ and $\ell_{\text{late}} = 24$ (75% depth), running inference in float16 on a single NVIDIA RTX A6000 GPU.

**Data.** We use 150 Wikipedia articles from the FOOTPRINTS dataset Feucht et al. [2024], each truncated to the first 256 tokens, yielding approximately 38,400 tokens total. We extract all contiguous spans of 2–6 tokens from each document.

**Baselines.** We compare against two baselines: (1) **spaCy NER**: named entities detected by the spaCy NER pipeline, providing an external reference for entity-type implicit vocabulary items; (2) **Low-score baseline**: 200 items with combined scores in the bottom quartile, classified with the same GPT-4.1 pipeline to establish the compositionality of spans that the model does *not* treat as units.

**Evaluation.** We report Spearman rank correlation between combined scores and compositionality ratings, Mann-Whitney $U$ tests comparing high- and low-scoring items, and chi-squared tests for category distributions. We use $\alpha = 0.05$ for all tests and report effect sizes (Cohen's $d$) where applicable.

## 4 Results

### 4.1 Extraction Statistics

From 150 Wikipedia articles, our probe-free method extracts 7,405 multi-token segment instances (6,494 unique) through greedy segmentation. The mean combined score is 0.532 (median 0.591), with scores spanning the full $[0, 1]$ range. Figure 1 shows the distribution of combined scores and its component signals. As a reference, spaCy NER detects 3,185 named entities in the same corpus, and 76.1% of our extracted items overlap with at least one NER entity.

### 4.2 Compositionality Analysis

**Category distribution.** Table 1 shows the classification results for 500 stratified items. Fragments—meaningless token spans such as partial words split across BPE boundaries (e.g., "arus which

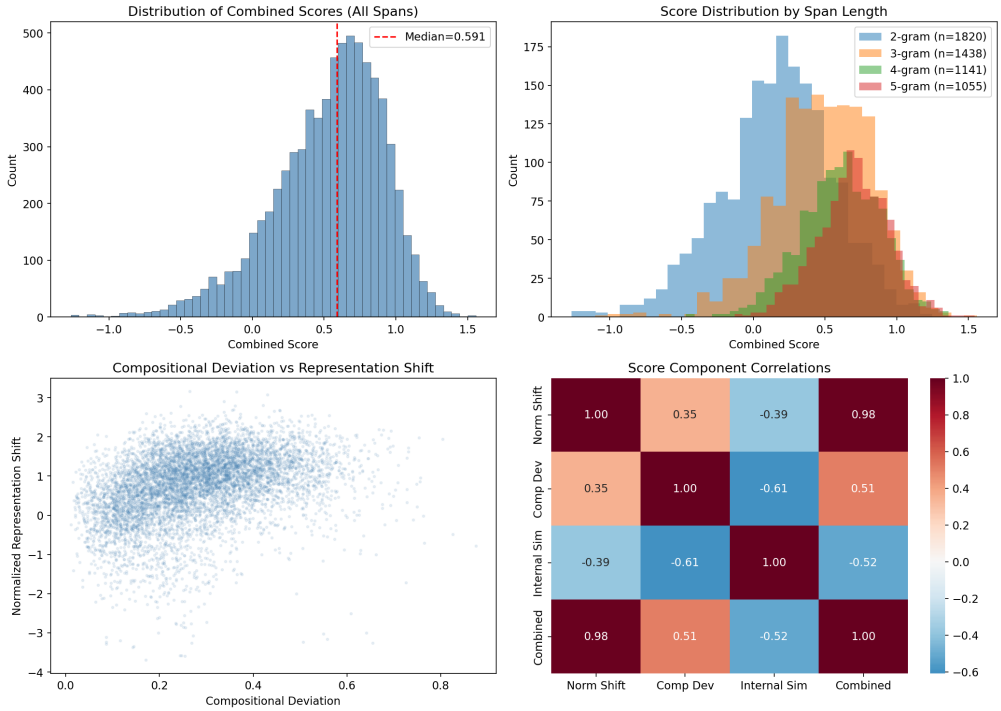


Figure 1: Distribution of combined scores and component signals. *(Left)*Combined score distribution across all extracted items. *(Right)*Relationship between compositional deviation and representation shift, the two primary components of the combined score.

Table 1: Classification of 500 stratified implicit vocabulary items by GPT-4.1. We report counts, percentages, mean compositionality scores (0 = fully non-compositional, 1 = fully compositional), and mean erasure-like scores. Best compositionality discrimination in **bold**.

| Category | Count | % | Comp. | Score |
|---|---|---|---|---|
| Fragment | 240 | 48.0 | 0.327 | 0.531 |
| Named entity | 99 | 19.8 | **0.379** | 0.623 |
| Compositional phrase | 74 | 14.8 | 1.000 | 0.586 |
| Technical term | 40 | 8.0 | 0.740 | **0.733** |
| Fixed expression | 30 | 6.0 | 0.730 | 0.480 |
| Compound word | 15 | 3.0 | 0.687 | 0.641 |
| Idiom | 2 | 0.4 | 0.300 | 0.339 |

forbids")—account for 48% of items, indicating that the probe-free method has higher noise than probe-based erasure. Among the 260 meaningful items, named entities are the largest category (38.1%), followed by compositional phrases (28.5%), technical terms (15.4%), fixed expressions (11.5%), compound words (5.8%), and idioms (0.8%).

**Score–compositionality correlation.** Among the 260 meaningful items (excluding fragments), we find a significant negative correlation between the combined score and compositionality (Spearman $\rho = -0.182$, $p = 0.003$). Figure 2 visualizes this relationship, showing that higher-scoring items cluster toward lower compositionality values.

**High vs. low score comparison.** Splitting meaningful items at the median combined score, high-scoring items have a mean compositionality of 0.557 compared to 0.781 for low-scoring items (Mann-Whitney $U = 6{,}117.5$, $p < 0.0001$, Cohen's $d = -0.60$). This medium effect size confirms that the model's internal representation shifts preferentially target non-compositional sequences.

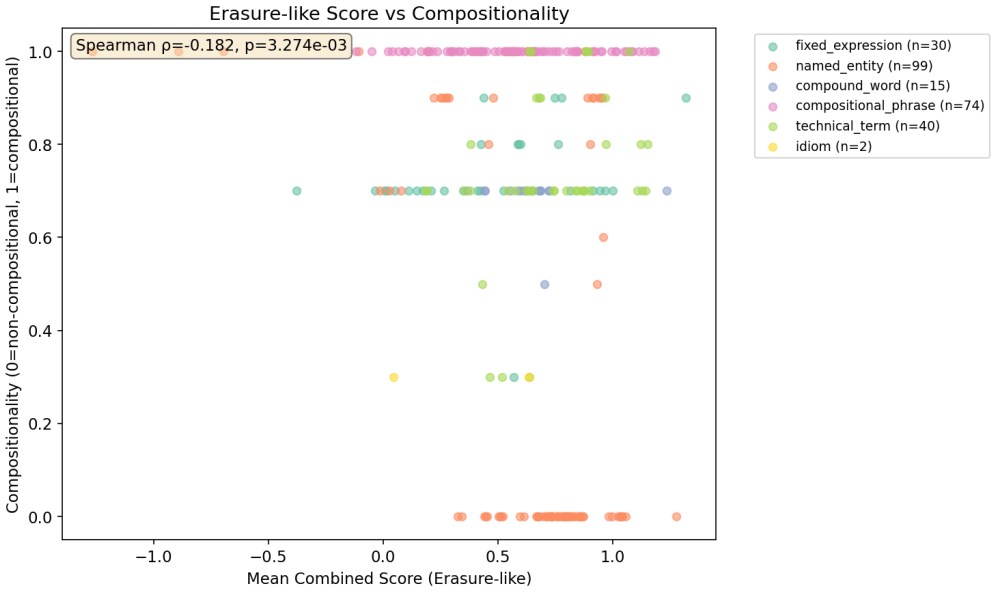


Figure 2: Compositionality rating vs. erasure-like score for 260 meaningful implicit vocabulary items, colored by category. The negative correlation (Spearman $\rho = -0.182$, $p = 0.003$) confirms that items the model treats as unified units tend to be less compositional.

Table 2: Statistical tests. All tests use $\alpha = 0.05$.

| **Test** | **Statistic** | $p$**-value** | **Effect size** |
|---|---|---|---|
| Score–compositionality correlation | Spearman $\rho = -0.182$ | 0.003 | Small–medium |
| High vs. low score compositionality | $U = 6{,}117.5$ | $< 0.0001$ | $d = -0.60$ (medium) |
| Non-comp. vs. comp. categories | $U = 999.0$ | $< 10^{-27}$ | $d = -1.72$ (large) |
| Vocabulary vs. baseline compositionality | $U = 50{,}637.0$ | 0.781 | n.s. |
| Category distribution vs. uniform | $\chi^2 = 155.5$ | $< 10^{-31}$ | — |

### 4.3 Statistical Tests

Table 2 summarizes our statistical tests. The core finding—that higher erasure-like scores predict lower compositionality—is robust across multiple tests. The category distribution is highly non-uniform ($\chi^2 = 155.5$, $p < 10^{-31}$), dominated by named entities. Notably, the overall compositionality of extracted items does *not* differ significantly from low-score baseline items ($U = 50{,}637$, $p = 0.781$), because the extraction process captures both compositional and non-compositional items; the discrimination occurs *within* the score distribution.

### 4.4 Category-Level Analysis

Figure 3 shows compositionality and score distributions by category. Named entities have the lowest mean compositionality (0.379) among meaningful categories, consistent with their non-compositional nature: "CompuServe" cannot be derived from "Compu" and "Serve." Technical terms show the highest mean erasure-like score (0.733), suggesting particularly strong representation merging for domain-specific terminology. Fixed expressions and compound words occupy a middle ground (compositionality 0.687–0.730), reflecting their partially compositional nature.

### 4.5 NER Overlap Analysis

Figure 4 shows the overlap between our extracted implicit vocabulary and spaCy NER entities. Of the 6,494 unique extracted items, 76.1% overlap with at least one NER entity, confirming that named entities are heavily represented in the implicit vocabulary. NER-matching items show slightly higher

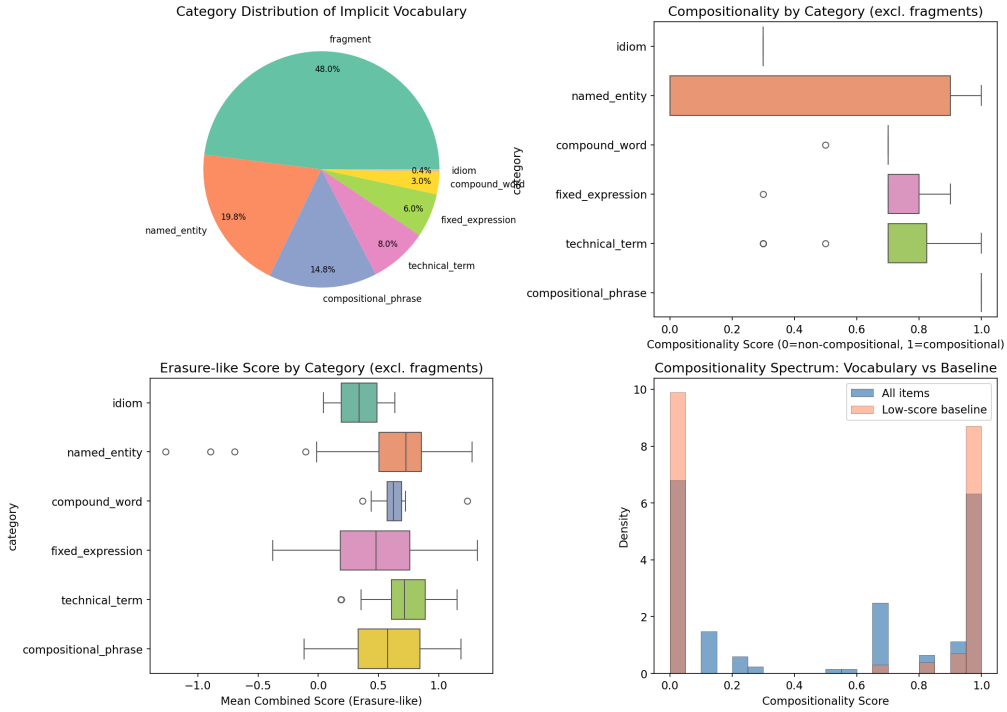


Figure 3: Category-level analysis of implicit vocabulary items. The figure shows the distribution of categories, compositionality scores by category, erasure-like scores by category, and the compositionality spectrum comparing vocabulary items to the low-score baseline. Named entities and idioms cluster at low compositionality, while compositional phrases are at 1.0 by definition.

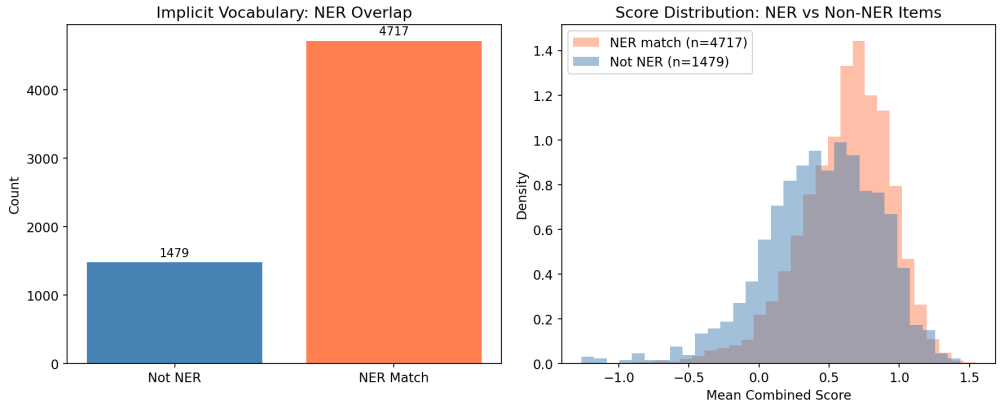


Figure 4: NER overlap analysis. 76.1% of extracted implicit vocabulary items overlap with spaCy NER entities. NER-matching items show a rightward shift in the score distribution, indicating slightly higher erasure-like scores.

erasure-like scores than non-NER items, consistent with our finding that the model treats named entities with strong merging behavior.

### 4.6 Vocabulary Size

Figure 5 shows how the size of the implicit vocabulary changes as a function of the combined score threshold. At no threshold, all 6,494 unique items are included. Raising the threshold progressively filters out lower-scoring items, yielding a more selective vocabulary dominated by named entities

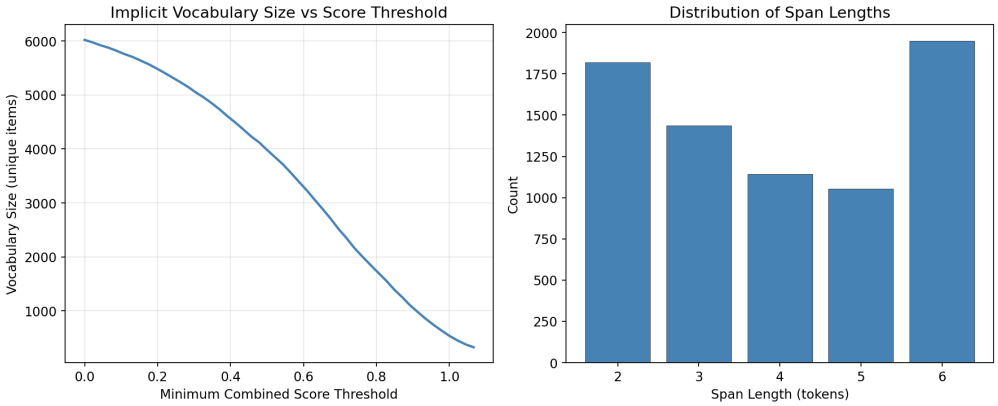


Figure 5: Vocabulary size as a function of combined score threshold (left) and distribution of span lengths among extracted items (right). Most implicit vocabulary items consist of 2–3 tokens.

and technical terms. The distribution of span lengths peaks at 2–3 tokens, with longer spans (5–6 tokens) being rarer.

## 5 Discussion

**What does the implicit vocabulary contain?** Our results paint a clear picture: the implicit vocabulary of MISTRAL-7B is dominated by named entities (38% of meaningful items), followed by technical terms (15%) and fixed expressions (12%). These categories share a common property—their meaning cannot be straightforwardly derived from their constituent tokens. "CompuServe" is not a composition of "Compu" and "Serve"; "quantum mechanics" carries meaning beyond the sum of "quantum" and "mechanics." The model appears to learn, through pre-training, that these sequences should be represented as unified semantic units.

**The compositionality spectrum.** A key finding is that compositionality is not binary. Rather than a clean division between "vocabulary items" and "non-vocabulary items," we observe a continuous spectrum. Named entities and idioms cluster at the low-compositionality end, compositional phrases at the high end, and technical terms, fixed expressions, and compound words span the middle. This suggests that the model develops graded representations of "unithood," treating some sequences as more cohesive than others. This finding echoes theoretical proposals in construction grammar Goldberg [2006] that linguistic units exist on a continuum of compositionality.

**Why do technical terms score highest?** An unexpected finding is that technical terms show the highest mean erasure-like scores (0.733), exceeding even named entities (0.623). We hypothesize that this reflects the model's need to perform particularly strong representational merging for domain-specific terminology, which often carries precise meanings that differ from the compositional reading of their parts. "Machine learning" in a technical context refers to a specific field, not literally to machines that learn. The model may allocate more representational resources to resolving this ambiguity.

**Fragment rate and method limitations.** The 48% fragment rate is the primary limitation of our probe-free method. Many fragments are partial words split across BPE boundaries (e.g., "icentennial" from "Bicentennial") or semantically incoherent token sequences (e.g., "arus which forbids"). The greedy segmentation algorithm assigns every token to some span, so low-information positions inevitably produce meaningless segments. The probe-based method of Feucht et al. [2024] achieves higher precision because probe accuracy provides a direct readout of whether the model has "erased" token-level information, while our combined score is a noisier proxy. However, the relative ordering of items by our score is informative: high-scoring items are significantly less compositional than low-scoring items, even if the absolute scores contain more noise.

**Implications for tokenizer design.** Our findings suggest that models would benefit from tokenizers that incorporate frequent non-compositional multi-token sequences as explicit vocabulary items. The 99 named entities, 40 technical terms, and 30 fixed expressions identified in our sample of 500

items represent cases where the model must use its limited representational capacity to merge tokens that could have been pre-merged by the tokenizer. This aligns with the proposal of Yang [2024] to learn tokenizer vocabularies that include multiword expressions.

**Limitations.** Beyond the fragment rate, we note several limitations. First, we analyze only MISTRAL-7B; cross-model validation with architectures of different sizes and vocabulary sizes would strengthen the findings. Second, our compositionality "ground truth" comes from GPT-4.1 rather than human annotators, introducing potential biases from one LLM judging another's representations. Third, the Wikipedia corpus overrepresents named entities and underrepresents colloquial idioms; results may differ substantially for conversational text. Fourth, we process only the first 256 tokens per document, missing items that appear later in longer articles. Finally, the combined score weights (0.4/0.4/0.2) were chosen heuristically; a principled optimization of these weights could improve precision.

## 6 Conclusion

We introduced a probe-free method for extracting and characterizing the implicit vocabulary of large language models. By combining representation shift, compositional deviation, and internal similarity signals from model hidden states, our method generalizes to any transformer without requiring pre-trained probes. Applied to MISTRAL-7B on 150 Wikipedia articles, we extracted 6,494 unique multi-token implicit vocabulary items and showed that items with higher erasure-like scores are significantly less compositional (Spearman $\rho = -0.182$, $p = 0.003$; Cohen's $d = -0.60$). Named entities constitute the largest non-compositional category (38% of meaningful items), followed by technical terms (15%) and fixed expressions (12%), and the compositionality distribution is continuous rather than binary.

These findings demonstrate that LLMs develop vocabulary-like internal representations that extend far beyond their explicit tokenizer vocabulary, particularly for named entities and conventionalized phrases. Future work should validate these findings across model families and vocabulary sizes, replace LLM-based compositionality judgments with human annotations, and explore whether the implicit vocabulary can directly inform tokenizer design for improved language modeling.

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